

Chapter 4

TRIGONOMETRY

Real-life Motivation

Trigonometry is used everywhere around us: measuring heights of buildings, calculating distances of stars and planets, designing bridges, analysing sound waves, and studying periodic motion. The study of trigonometry helps us understand the relationship between angles and sides of triangles and forms the foundation of mathematics, physics, engineering, and astronomy.

4.1 Introduction

Trigonometry

The word **trigonometry** is derived from the Greek words *trigon* (triangle) and *metron* (measure). It deals with the study of relationships between angles and sides of triangles.

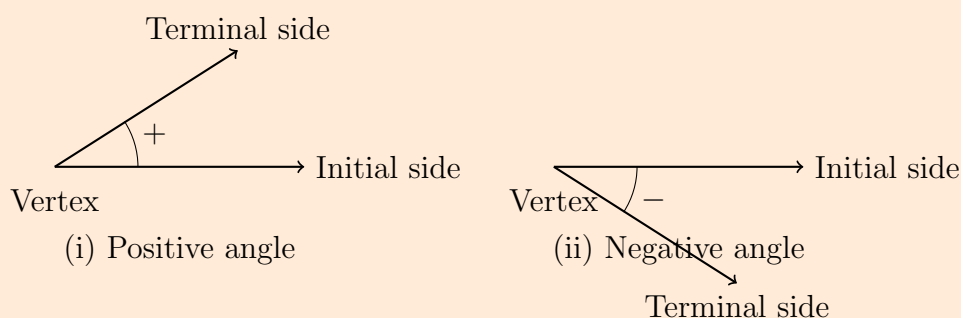
4.2 Angles

Angle

An angle is a measure of rotation of a ray about its initial point.

- Fixed point → **Vertex**
- Starting ray → **Initial side**
- Final ray → **Terminal side**

Positive and Negative Angles



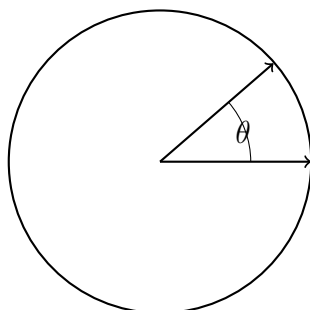
Anticlockwise rotation gives a positive angle, while clockwise rotation gives a negative angle.

4.2.1 Degree Measure

Degree Measure

If a rotation is $\frac{1}{360}$ of a complete revolution, the angle is said to have a measure of **one degree** (1°).

$$1^\circ = 60', \quad 1' = 60''$$



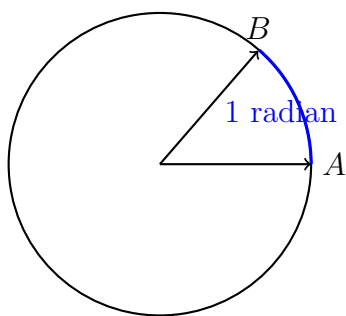
Angle in degree measure

4.2.2 Radian Measure

Radian Measure

An angle subtended at the centre of a circle by an arc whose length is equal to the radius of the circle is called **one radian**.

$$\theta = \frac{l}{r}, \quad l = r\theta$$



Arc length = radius

One complete revolution corresponds to 2π radians.

4.2.3 Relation Between Degree and Radian

Important Relation

$$180^\circ = \pi \text{ radians}$$

$$1^\circ = \frac{\pi}{180} \text{ radian,} \quad 1 \text{ radian} = \frac{180^\circ}{\pi}$$

Common Angle Values

Degree	30°	45°	60°	90°	180°
Radian	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π

4.3 Trigonometric Functions

In earlier classes, we studied trigonometric ratios of acute angles as the ratio of the sides of a right-angled triangle. We now extend these definitions to **angles of any magnitude** using the concept of the **unit circle**.

Unit Circle

Unit Circle

A circle with centre at the origin and radius equal to one unit is called the **unit circle**.

Let $P(a, b)$ be any point on the unit circle and $\angle AOP = x$ radians, measured anti-clockwise from the positive x -axis.

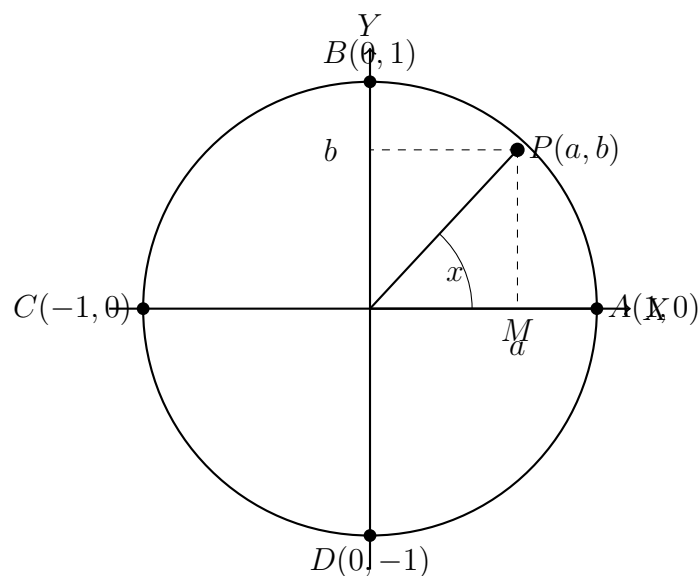


Fig. 4.6 : Unit Circle

Definition of Sine and Cosine

Definitions

If $P(a, b)$ is a point on the unit circle corresponding to angle x , then

$$\cos x = a, \quad \sin x = b$$

Since $\triangle OMP$ is a right-angled triangle,

$$OM^2 + MP^2 = OP^2 \Rightarrow a^2 + b^2 = 1$$

Hence, for every point on the unit circle,

$$\sin^2 x + \cos^2 x = 1$$

Quadrantal Angles

Quadrantal Angles

Angles which are integral multiples of $\frac{\pi}{2}$ are called **quadrantal angles**.

$$0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$$

For these angles:

Angle	$\cos x$	$\sin x$
0	1	0
$\frac{\pi}{2}$	0	1
π	-1	0
$\frac{3\pi}{2}$	0	-1
2π	1	0

Periodicity of Sine and Cosine

If we take one complete revolution from a point on the unit circle, we return to the same point.

Periodic Nature

$$\sin(x + 2\pi) = \sin x, \quad \cos(x + 2\pi) = \cos x$$

Also,

$$\begin{aligned} \sin x = 0 &\Rightarrow x = n\pi \\ \cos x = 0 &\Rightarrow x = (2n + 1)\frac{\pi}{2}, \quad n \in \mathbb{Z} \end{aligned}$$

Other Trigonometric Functions

Definitions of Other Functions

$$\begin{aligned} \csc x &= \frac{1}{\sin x}, \quad x \neq n\pi \\ \sec x &= \frac{1}{\cos x}, \quad x \neq (2n + 1)\frac{\pi}{2} \\ \tan x &= \frac{\sin x}{\cos x}, \quad x \neq (2n + 1)\frac{\pi}{2} \\ \cot x &= \frac{\cos x}{\sin x}, \quad x \neq n\pi \end{aligned}$$

Important Identities

$$\sin^2 x + \cos^2 x = 1$$

It follows that:

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

Values of Trigonometric Functions

Values of Trigonometric Functions

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	2π
$\sin x$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	0
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	-1	1
$\tan x$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	ND	0	0

ND = Not Defined

4.3.1 Sign of Trigonometric Functions

Let $P(a, b)$ be a point on the unit circle corresponding to angle x .

Even and Odd Properties

$$\cos(-x) = \cos x, \quad \sin(-x) = -\sin x$$

Since $-1 \leq a \leq 1$ and $-1 \leq b \leq 1$, we have

$$-1 \leq \cos x \leq 1, \quad -1 \leq \sin x \leq 1.$$

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Signs in Different Quadrants

Quadrant-wise Signs

Function	I	II	III	IV
$\sin x$	+	+	−	−
$\cos x$	+	−	−	+
$\tan x$	+	−	+	−
$\csc x$	+	+	−	−
$\sec x$	+	−	−	+
$\cot x$	+	−	+	−

4.3.2 Domain and Range of Trigonometric Functions

From the definition of sine and cosine functions, we observe that they are defined for all real numbers.

Basic Result

$$-1 \leq \sin x \leq 1, \quad -1 \leq \cos x \leq 1$$

Domain and Range:

Domain and Range

Function	Domain	Range
$\sin x$	$\mathbb{R}$	$[-1, 1]$
$\cos x$	$\mathbb{R}$	$[-1, 1]$
$\tan x$	$x \neq (2n + 1)\frac{\pi}{2}$	$\mathbb{R}$
$\cot x$	$x \neq n\pi$	$\mathbb{R}$
$\sec x$	$x \neq (2n + 1)\frac{\pi}{2}$	$(-\infty, -1] \cup [1, \infty)$
$\csc x$	$x \neq n\pi$	$(-\infty, -1] \cup [1, \infty)$

where $n \in \mathbb{Z}$.

Behaviour of Trigonometric Functions:

Behaviour in Quadrants

The behaviour of trigonometric functions depends on the quadrant in which the angle lies.

Function	I Quadrant	II Quadrant	III Quadrant	IV Quadrant
$\sin x$	increases $0 \rightarrow 1$	decreases $1 \rightarrow 0$	decreases $0 \rightarrow -1$	increases $-1 \rightarrow 0$
$\cos x$	decreases $1 \rightarrow 0$	decreases $0 \rightarrow -1$	increases $-1 \rightarrow 0$	increases $0 \rightarrow 1$
$\tan x$	$0 \rightarrow \infty$	$-\infty \rightarrow 0$	$0 \rightarrow \infty$	$-\infty \rightarrow 0$
$\cot x$	$\infty \rightarrow 0$	$0 \rightarrow -\infty$	$-\infty \rightarrow 0$	$0 \rightarrow \infty$
$\sec x$	$1 \rightarrow \infty$	$-\infty \rightarrow -1$	$-\infty \rightarrow -1$	$1 \rightarrow \infty$
$\csc x$	$1 \rightarrow \infty$	$1 \rightarrow \infty$	$-\infty \rightarrow -1$	$-\infty \rightarrow -1$

Remark

The symbols ∞ and $-\infty$ indicate unbounded behaviour and not actual values.

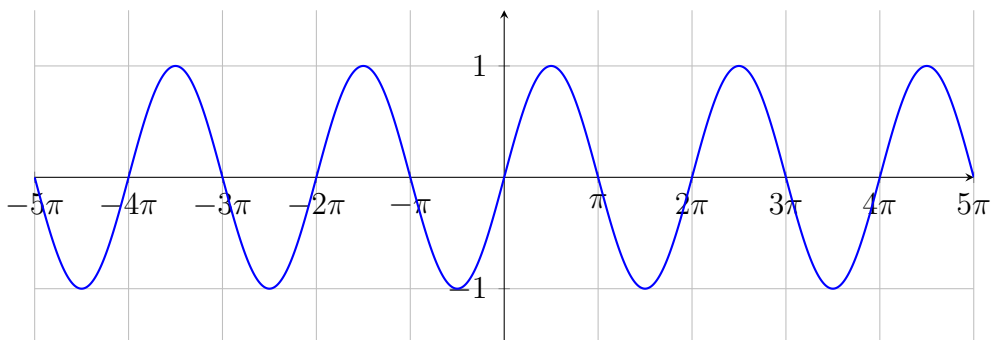
Periodicity:

Periodicity

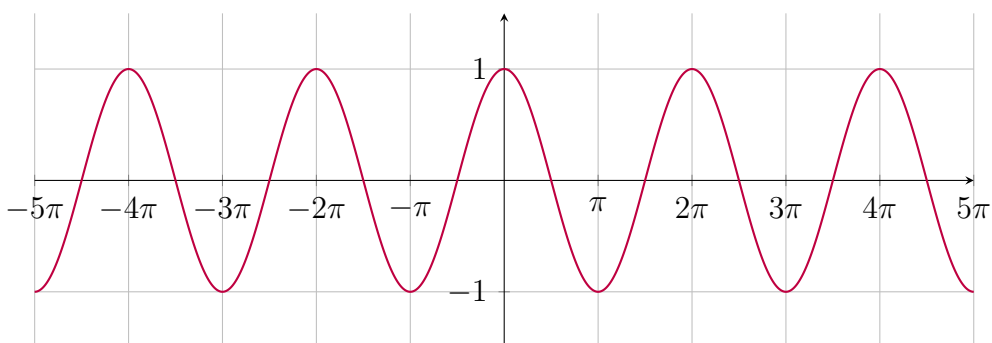
$$\sin(x + 2\pi) = \sin x, \quad \cos(x + 2\pi) = \cos x$$

$$\tan(x + \pi) = \tan x, \quad \cot(x + \pi) = \cot x$$

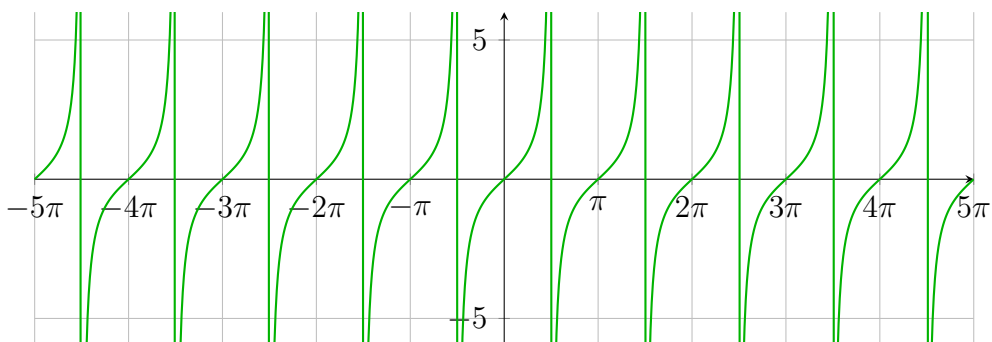
$$y = \sin x$$



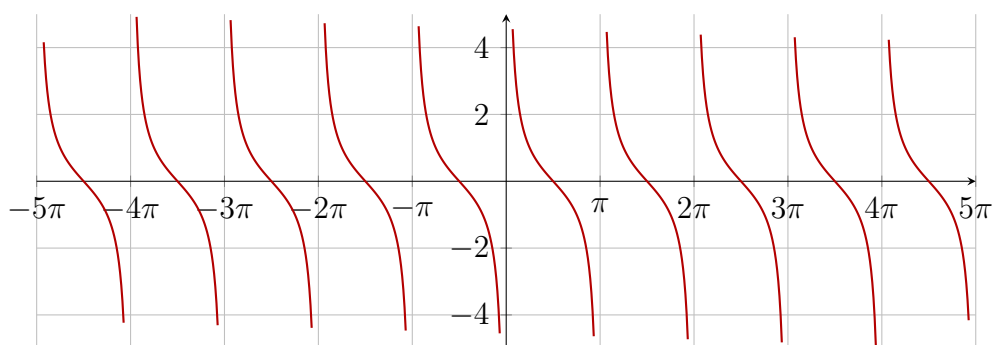
$$y = \cos x$$



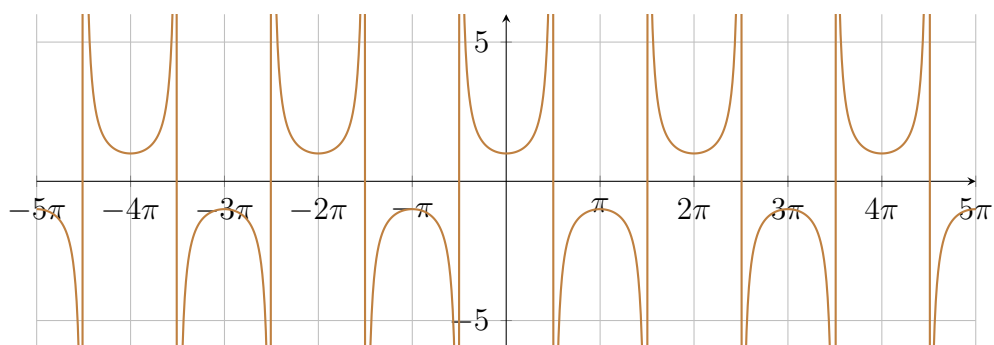
$$y = \tan x$$



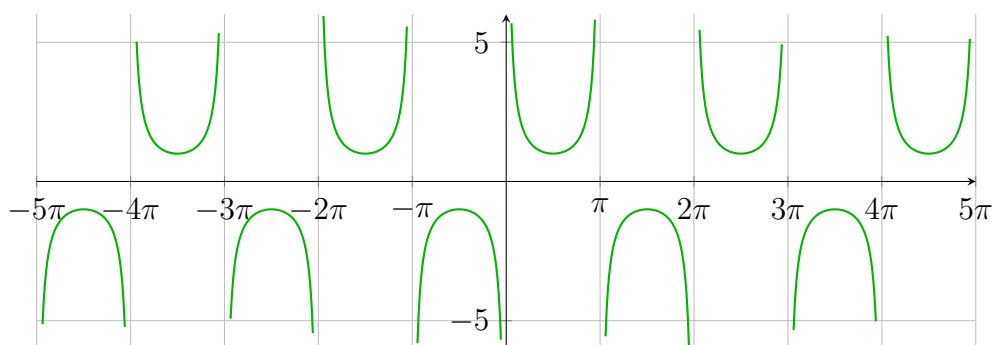
$$y = \cot x$$



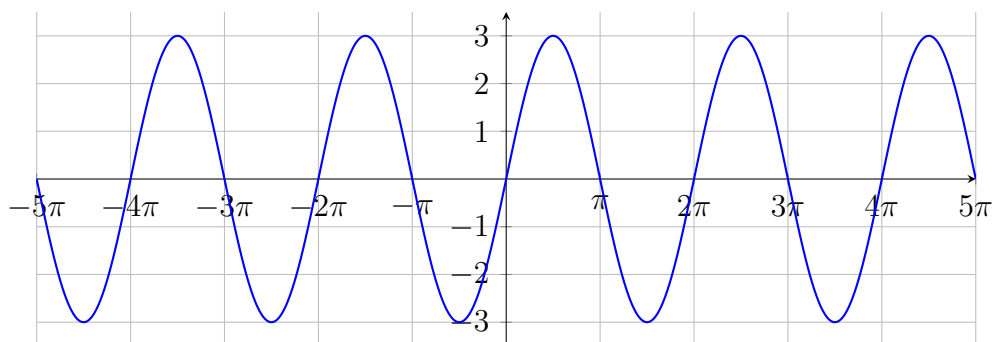
$$y = \sec x$$



$$y = \csc x$$



$$y = 3 \sin x$$



4.4 Trigonometric Functions of Sum and Difference of Two Angles

In this section, we derive expressions for trigonometric functions of the sum and difference of two angles. These results are called **trigonometric identities**.

Basic Identities

$$\sin(-x) = -\sin x, \quad \cos(-x) = \cos x$$

Cosine Identities

$$\cos(x + y) = \cos x \cos y - \sin x \sin y$$

$$\cos(x - y) = \cos x \cos y + \sin x \sin y$$

Sine Identities

$$\sin(x + y) = \sin x \cos y + \cos x \sin y$$

$$\sin(x - y) = \sin x \cos y - \cos x \sin y$$

Allied Angle Identities

$$\cos\left(\frac{\pi}{2} - x\right) = \sin x, \quad \sin\left(\frac{\pi}{2} - x\right) = \cos x$$

$$\cos(\pi - x) = -\cos x, \quad \sin(\pi - x) = \sin x$$

$$\cos(\pi + x) = -\cos x, \quad \sin(\pi + x) = -\sin x$$

$$\cos(2\pi - x) = \cos x, \quad \sin(2\pi - x) = -\sin x$$

Tangent and Cotangent Formulae

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

$$\tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

$$\cot(x + y) = \frac{\cot x \cot y - 1}{\cot x + \cot y}$$

$$\cot(x - y) = \frac{\cot x \cot y + 1}{\cot y - \cot x}$$

Double and Triple Angle Formulae

$$\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

$$\cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$$

$$\sin 2x = 2 \sin x \cos x = \frac{2 \tan x}{1 + \tan^2 x}$$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

$$\sin 3x = 3 \sin x - 4 \sin^3 x$$

$$\cos 3x = 4 \cos^3 x - 3 \cos x$$

4.4 Trigonometric Functions of Sum and Difference of Two Angles

Basic Results

1. $\sin(-x) = -\sin x$
2. $\cos(-x) = \cos x$

Sum and Difference Formulae

$$\cos(x + y) = \cos x \cos y - \sin x \sin y$$

$$\cos(x - y) = \cos x \cos y + \sin x \sin y$$

$$\sin(x + y) = \sin x \cos y + \cos x \sin y$$

$$\sin(x - y) = \sin x \cos y - \cos x \sin y$$

Tangent Formulae

If none of the angles is an odd multiple of $\frac{\pi}{2}$, then

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}, \quad \tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}.$$

Multiple Angle Formulae

Double Angle Formulae

$$\sin 2x = 2 \sin x \cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

Triple Angle Formulae

$$\sin 3x = 3 \sin x - 4 \sin^3 x$$

$$\cos 3x = 4 \cos^3 x - 3 \cos x$$

$$\tan 3x = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$$

Sum-to-Product Formulae

Sum and Difference Conversion

$$\cos x + \cos y = 2 \cos \frac{x+y}{2} \cos \frac{x-y}{2}$$

$$\cos x - \cos y = -2 \sin \frac{x+y}{2} \sin \frac{x-y}{2}$$

$$\sin x + \sin y = 2 \sin \frac{x+y}{2} \cos \frac{x-y}{2}$$

$$\sin x - \sin y = 2 \cos \frac{x+y}{2} \sin \frac{x-y}{2}$$

Examples

Example 1

Prove that:

$$3 \sin \frac{\pi}{6} \sec \frac{\pi}{3} - 4 \sin \frac{5\pi}{6} \cot \frac{\pi}{4} = 1$$

Solution:

$$= 3 \cdot \frac{1}{2} \cdot 2 - 4 \cdot \frac{1}{2} \cdot 1 = 3 - 2 = 1$$

Example 2

Find the value of $\sin 15^\circ$.

Solution:

$$\begin{aligned}\sin(45^\circ - 30^\circ) &= \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ \\ &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{\sqrt{3} - 1}{2\sqrt{2}}\end{aligned}$$

Example 3

Find the value of $\sin 30^\circ \cos 60^\circ + \cos 30^\circ \sin 60^\circ$.

Solution:

$$= \sin(30^\circ + 60^\circ) = \sin 90^\circ = 1$$

Example 4

Prove that $\sin x(1 + \tan^2 x) = \sec x \tan x$.

Solution:

$$\sin x(1 + \tan^2 x) = \sin x \sec^2 x = \frac{\sin x}{\cos^2 x} = \sec x \tan x$$

Example 5

Find $\cos(-210^\circ)$.

Solution:

$$\cos(-210^\circ) = \cos 210^\circ = -\cos 30^\circ = -\frac{\sqrt{3}}{2}$$

Example 6

If $\sin x = \frac{3}{5}$ and x lies in the first quadrant, find $\cos x$.

Solution:

$$\cos x = \sqrt{1 - \sin^2 x} = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$$

Example 7

Prove that $\frac{1 - \tan^2 x}{1 + \tan^2 x} = \cos 2x$.

Solution:

$$\frac{1 - \tan^2 x}{1 + \tan^2 x} = \frac{\cos^2 x - \sin^2 x}{\cos^2 x + \sin^2 x} = \cos 2x$$

Example 8

Find the value of $\sin 75^\circ$.

Solution:

$$\sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$$

Example 9

Evaluate $\tan 225^\circ$.

Solution:

$$\tan(180^\circ + 45^\circ) = \tan 45^\circ = 1$$

Example 10

Prove that $\sin 3x = 3 \sin x - 4 \sin^3 x$.

Solution:

$$\begin{aligned}\sin(2x + x) &= \sin 2x \cos x + \cos 2x \sin x \\ &= 2 \sin x \cos^2 x + (1 - 2 \sin^2 x) \sin x = 3 \sin x - 4 \sin^3 x\end{aligned}$$

Example 11

Find the value of $\cos\left(\frac{\pi}{2} - x\right)$.

Solution:

$$\cos\left(\frac{\pi}{2} - x\right) = \sin x$$

Example 12

If $\tan x = \frac{5}{12}$, find $\sin x$ and $\cos x$.

Solution:

$$\sin x = \frac{5}{13}, \quad \cos x = \frac{12}{13}$$

Example 13

Prove that $\sin^2 x + \cos^2 x = 1$.

Solution: Divide numerator and denominator of $\tan^2 x + 1 = \sec^2 x$ by $\sec^2 x$.

$$\sin^2 x + \cos^2 x = 1$$

Example 14

Evaluate $\tan\left(-\frac{\pi}{4}\right)$.

Solution:

$$\tan(-x) = -\tan x \Rightarrow \tan\left(-\frac{\pi}{4}\right) = -1$$

Example 15

Find $\cos 3x$ in terms of $\cos x$.

Solution:

$$\cos 3x = 4 \cos^3 x - 3 \cos x$$

Example 16

Prove that $\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$.

Solution:

$$\tan(x + y) = \frac{\sin(x + y)}{\cos(x + y)} = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

Example 17

Evaluate $\sin 660^\circ$.

Solution:

$$\sin 660^\circ = \sin(660^\circ - 360^\circ) = \sin 300^\circ = -\frac{\sqrt{3}}{2}$$

Exercise — Trigonometry

Exercise A

Q.1. Evaluate $\sin 75^\circ$.

Q.2. Find the value of $\cos 210^\circ$.

Q.3. If $\sin x = \frac{5}{13}$ and x lies in the first quadrant, find $\cos x$.

Q.4. Prove that:

$$(1 + \tan^2 x)(1 - \sin^2 x) = 1$$

Q.5. Evaluate $\tan\left(-\frac{\pi}{3}\right)$.

Q.6. Find the value of $\sin\left(\frac{\pi}{2} - 30^\circ\right)$.

Q.7. If $\tan x = \frac{3}{4}$ and x lies in the third quadrant, find $\sin x$.

Q.8. Prove that:

$$\frac{1 - \cos 2x}{1 + \cos 2x} = \tan^2 x$$

Q.9. Find the value of $\cos\left(\frac{7\pi}{6}\right)$.

Q.10. Evaluate $\sin 660^\circ$.

Q.11. Prove that:

$$\sin(x + y) + \sin(x - y) = 2 \sin x \cos y$$

Q.12. Find $\cos 3x$ in terms of $\cos x$.

Q.13. If $\sec x = \frac{13}{5}$ and x lies in the fourth quadrant, find $\tan x$.

Q.14. Evaluate $\tan 225^\circ$.

Q.15. Prove that:

$$\cos x + \cos y = 2 \cos \frac{x + y}{2} \cos \frac{x - y}{2}$$

Answers:

Answers

Ans.1. $\frac{\sqrt{6} + \sqrt{2}}{4}$

Ans.2. $-\frac{\sqrt{3}}{2}$

Ans.3. $\frac{12}{13}$

Ans.4. L.H.S. = R.H.S.

Ans.5. $-\sqrt{3}$

Ans.6. $\frac{1}{2}$

Ans.7. $-\frac{3}{5}$

Ans.8. L.H.S. = R.H.S.

Ans.9. $-\frac{\sqrt{3}}{2}$

Ans.10. $-\frac{\sqrt{3}}{2}$

Ans.11. $2 \sin x \cos y$

Ans.12. $4 \cos^3 x - 3 \cos x$

Ans.13. $-\frac{12}{5}$

Ans.14. 1

Ans.15. $2 \cos \frac{x+y}{2} \cos \frac{x-y}{2}$